Lecture 11 — Neighbors practice

Monday, September 28

Last updated: October 12, 2026.

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

In class Monday, September 28. Quiz 4 was Friday, September 25. Midterm 1 is Friday, October 2 — prep materials land on Lecture 15 Wednesday.

This meeting is **software practice** on angle, nearest neighbors, and clustering from Lectures 08–10. It is **not** a graded lab (Labs 1–6 begin Friday, October 16). Bring a laptop if you have one.

Work in Octave or Python. Use the syntax reference and Lachniet Ch. 1 as needed.

Goals for today

1. Compute a few distances between feature vectors.
2. Name a nearest neighbor and say what it means in context.
3. Notice how rescaling one feature can change who looks "closest."

Warm-up ideas

Pick three short feature vectors from class examples (or invent song / sensor features). Compute pairwise Euclidean distances, identify the nearest neighbor of a query, then multiply one coordinate by 10 and repeat.

Related: Lectures · Schedule · Lab 1 (graded version later).